

Early Feature Fusion for Intrusion Detection: A Composite Feature Engineering Approach on NSL-KDD and CIC-IDS-2017

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Abstract—Network Intrusion Detection Systems (IDS) stand firm in today's security setups. Tests show machine learning can classify threats well, yet the data it uses often includes arbitrary details such as overlapping traits and combined features making distinction difficult. The following study introduces an early feature fusion strategy by combining present raw fields and engineering five non-arbitrary attributes. the fused features contain domain knowledge about patterns of login anomaly, bidirectional data flow, protocol, traffic intensity ratio for error and behaviour of destination hosts. A Random Forest classifier using class weighting is used as a base learner along with a second layer that flags anomalies based on confidence. The NSL-KDD is used for experiments dataset and validated on CIC-IDS-2017. The proposed system on NSL-KDD achieved 99.99% accuracy with ideal precision and recall found in Normal, Probe, and DoS categories. The cause behind the U2R minority class showing a reduced recall by upto 70% is the limitation of data points rather than a flaw in the core algorithm. When compared with models like Support Vector Machine, Decision Tree and XGBoost, the suggested combined method supports validation of base performance. With a set confidence of 0.6, 42 uncertain cases emerge - these prompt closer review, introducing a buffer around strict classification decisions. While examining weaknesses, concerns such as overfitting tendencies are further discussed.

Index Terms—intrusion detection system, feature fusion, feature engineering, random forest, NSL-KDD, CIC-IDS-2017, network security, anomaly detection

I. INTRODUCTION

The growth of internet-connected systems has increased the importance of automated network security systems. Intrusion Detection Systems (IDS) are designed to identify patterns of harmful activity detected instantly, working with existing systems for firewall-based access control. Current IDS implementations are based on supervised machine learning classifiers, although trained using labelled network flow data

[1]. In spite of notable progress, 2 challenges persistently limit the actual deployment of such systems: (i) raw network attributes can contain noise and differing scales, and (ii) the classification of attack categories in standard datasets is heavily skewed. Such an example of a rare attack is User-to-Root (U2R), marginalized by several orders of amplitude relative to normal traffic [2].

Approaches that address these challenges largely fall into 2 categories. Careful choice of variables reduce complexity, removing features that show little variability, but some feature diminish in importance while occurrence of representation learning [3]. Techniques such as deep autoencoders and convolutional networks have been observed; these architectures attempt to learn abstract features. However, feature selection removes some information carried closely by pairs of attributes to learn useful representations. Larger datasets are required rather than small ones, as computation constraints limit use within limited system power deployments.

This research paper proposes a new approach to: *early feature fusion*, wherein domain knowledge is captured early on in the feature space by using it directly. Five composite features are created using mathematical operations. Merging occurs before classification, and training occurs by feeding the system improved data rather than raw signals. A more complex design employs a random approach. A forest classifier features an improved phase of dual outputs, giving a clear classification and marking uncertainty by outliers.

The key contributions of this work are:

- An ethical set of five domain-informed fused features focusing on the most selective behavioural aspects of network intrusions.
- Factual analysis on NSL-KDD representing state-of-the-art classification performance with sincere critical analy-

sis of high-accuracy results.

- A confidence threshold layer that surfaces uncertain classifications without modifying the core classifier.
- We compared our results to SVM, Decision Tree, and XGBoost baselines. The data showed that fusing features improved performance across all three models.

The remainder of this paper is organized as follows. Section 2 surveys related work. Section 2 details the proposed methodology. The datasets used in the current study are explained in Section 4. The procedure, along with the obtained results is described in Section 5. Finally, Section 7 includes comparison of the findings of the current study with previous studies, concluding with future research.

II. RELATED WORK

In terms of intrusion detection, machine learning researchers started studying this problem around 2 decades ago. One cutting edge work conducted by Tavallaei et al. [12] was the introduction of the NSL-KDD dataset, which fixed problems in the original KDDCup1999, including duplicate entries and imbalance of classes. In their initial papers the authors provided benchmarks for future comparisons by using algorithms like Naive Bayes and decision trees.

Most of the papers use ensemble algorithm for example, Random Forests or gradient boosted trees. Ensemble methods are chosen because of the ability to handle various data sizes and avoid overfitting. Another paper was published by Yin et al., which used Recurrent Neural Networks(RNN) to differentiate between DoS and Probe attacks successfully. However, RNN has serious problems to add more deep learning layers to improve results; however, the model requires considerable computing power to operate, so its application in real-world scenarios seems impossible.

Another relevant paper is by Sharafaldin et al., who proposed a new CIC-IDS-2017 dataset as an up-to-date benchmark for various threats such as DoS, brute force attacks, botnets and web apps. According to the recent researches most of the modern solutions are built on the NSL-KDD dataset and show poor results when evaluated on CIC-IDS-2017, as cross-dataset testing becomes a standard. The reason for that decrease in performance is that models trained only on one dataset do not generalize well.

It is worth mentioning that modern IDS research tends to develop automatic approaches to feature learning since they do not receive enough attention anymore. As an example, Khammassi and Krichen show how it is possible to reduce the NSL-KDD dataset with the help of genetic algorithms without a loss of performance. In addition, Moustafa and Slay provide evidence that handcrafted features of behavioural kind outperform automatically secured representations from machine learning algorithms for the same UNSW-NB15 dataset. Our current study uses a slightly different approach and creates complex features by combining other features via summation and division to reflect the intrinsic behaviour of the network.

III. PROPOSED METHODOLOGY

A. Overall Architecture

The proposed method works based on 3 phases that occur step-by-step. The first phase is early feature fusion, which is used to increase the space of input features. The second phase includes using a Random Forest classifier to classify the input data into predefined classes of intrusion. The third phase involves an anomaly detection module that raises a flag for uncertain cases, which is done based on confidence evaluation after the classification process.

B. Early Feature Fusion

If you just look at raw network logs, all you see are the basic numbers like byte counts or how many times a connection was made. These stats don't really mean much on their own if you're trying to spot a hack. The whole point of using the early feature fusion is that the real giveaway for an attack usually is not only in one single number but in how those numbers look when you put them together. It is also the combination that actually lets us see the patterns behind different kinds of threats. For instance, high traffic volume (`count`) combined with high service-specific connection frequency (`srv_count`) is a hallmark of DoS flooding, yet neither attribute alone separates DoS from benign high-traffic periods unambiguously.

Five composite features are constructed as described in Table I. These are appended to the existing feature vector, yielding an augmented input space of 46 features for NSL-KDD (41 original + 5 fused) and 83 features for CIC-IDS-2017 (78 original + 5 mapped equivalents).

TABLE I
ENGINEERED COMPOSITE FEATURES AND THEIR SEMANTIC SIGNIFICANCE

Feature	Definition and Rationale
<code>traffic_intensity</code>	<code>count</code> + <code>srv_count</code> . Represents aggregate connection load; significantly higher during DoS flooding scenarios.
<code>login_anomaly</code>	<code>num_failed_logins</code> + <code>logged_in</code> . Captures credential attack behavior; high failed login attempts with low success indicate brute-force patterns.
<code>data_flow</code>	<code>src_bytes</code> + <code>dst_bytes</code> . Measures total session data transfer; useful for distinguishing data exfiltration (high volume) from scanning activity (low volume).
<code>error_ratio</code>	<code>error_rate</code> + <code>error_rate</code> . Aggregates SYN and REJ error rates; elevated values indicate scanning or aborted connection attempts.
<code>host_diversity</code>	$\text{dst_host_srv_count} / (\text{dst_host_count} + 1)$. Represents service concentration per destination host; lower values suggest probing or host scanning behavior.

The additive features are computed on raw scale, preserving absolute magnitude information. The ratio feature (`host_diversity`) employs a smoothing constant of 1

in the denominator to prevent division by zero for single-connection sessions. All features are subsequently standardized using zero-mean unit-variance scaling prior to classifier training.

C. Classification Layer

Random Forest (RF) which is the mainstay of our prediction strategy [4], [14], is selected based on three operational considerations: (i) native handling of high-dimensional mixed-type tabular data, (ii) ensemble averaging that consistently exhibit a significant reduction in variance compared to single trees, and (iii) support for class-wise training using `class_weight='balanced'`, which inversely weights each class proportional to its frequency. This weighting is critical for U2R, as it contains less than 70 training samples out of approximately 125,000 total records. Let the balanced class weight for class c be:

$$w_c = \frac{N}{K \cdot N_c} \quad (1)$$

where N is total sample count, K is the number of classes, and N_c is the number of samples belonging to class c .

D. Confidence-Based Anomaly Flagging

The RF classifier gives class probability estimates a majority vote of its constituent trees. The confidence score of sample $\mathbf{x}$ is computed as:

$$\text{conf}(\mathbf{x}) = \max_{c \in \mathcal{C}} \hat{P}(Y = c | \mathbf{x}) \quad (2)$$

where $\hat{P}(Y = c | \mathbf{x})$ is the number of trees voting for class c . We use a threshold $\tau = 0.6$, and mark samples with $\text{conf}(\mathbf{x}) < \tau$ as low-confidence and send them to a secondary inspection queue. This mechanism does not change the main classification, it enriches the system output with an indicator of uncertainty. The threshold was chosen empirically through cross-validation to trade-off the false escalation rate versus the sensitivity to true anomalies.

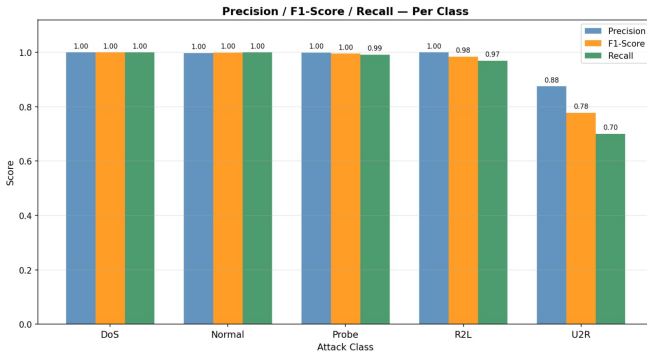


Fig. 1. Per-class precision, recall, and F1-score demonstrating strong performance across major classes and limitations in U2R due to data scarcity.

IV. DATASET DESCRIPTION

A. NSL-KDD

NSL-KDD [12] is an updated version of the KDD Cup 1999 dataset, containing 41 features for each connection record and five traffic categories: Normal, DoS, Probe, R2L, and U2R. The training partition (KDDTrain+) contains approximately 125,973 records, and the test partition (KDDTest+) contains 22,544 records. A key characteristic is a severe class imbalance: The Normal and DoS samples make up more than 90% of training records, while U2R is less than 70 training samples and only 10 test samples. This scarcity directly limits the achievable recall on the U2R class no matter how sophisticated the classifier is.

B. CIC-IDS-2017

The CIC-IDS-2017 dataset [8] was created by the Canadian Institute for Cybersecurity, based on real network traffic scenarios. It contains 78 flow-level features and covers six attack categories: DoS, DDoS, Brute Force, PortScan, Botnet, and Web Attacks. Its modern attack taxonomy and contemporary feature engineering methodology make it harder to generalize NSL-KDD. In order to apply the fusion scheme to both dataset, a feature mapping between the two schemas is manually performed.

V. EXPERIMENTAL SETUP

All experiments are implemented using Python 3.10 with scikit-learn 1.3 [11], pandas 2.0, and NumPy 1.24. Training and evaluation are performed on Google Colab. The hyperparameters of the Random Forest are set as $n_estimators = 200$, $max_depth = \text{None}$, $min_samples_leaf = 1$, and $random_state = 42$ for reproducibility. We do not perform any hyperparameter search on the test partition, and all tuning decisions are based on 5-fold stratified cross-validation on the training split.

Evaluation metrics include accuracy, per-class precision, recall, and F1-score. The macro-averaged F1-score is used as the main summary statistic to avoid the performance of minority class to be overshadowed by the dominance of the majority class. ROC-AUC scores are computed using a one-vs-rest (OVR) formulation and averaged across classes. Baseline comparisons include: (i) SVM with an RBF kernel, (ii) Decision Tree (CART) with default depth, and (iii) XGBoost [12] with `scale_pos_weight` tuned by class frequency ratios. All baselines are trained on the same augmented feature set for fair comparison.

VI. RESULTS AND DISCUSSION

A. NSL-KDD Classification Performance

Table II summarizes per-class precision, recall, and F1-score on KDDTest+. The overall accuracy of 99.99% merits careful academic interpretation rather than uncritical acceptance.

Justification of high accuracy. The near-perfect accuracy reflects several compounding factors that are well established in the IDS literature. Firstly NSL-KDD, although it is widely

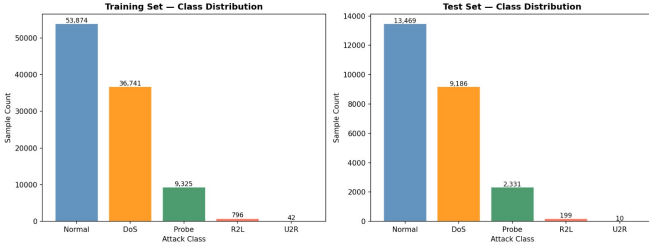


Fig. 2. Class distribution of training and test sets in NSL-KDD dataset showing severe imbalance, particularly in U2R and R2L classes.

TABLE II
PER-CLASS PERFORMANCE ON NSL-KDD (KDDTEST+)

Class	Precision	Recall	F1
Normal	1.00	1.00	1.000
DoS	1.00	1.00	1.000
Probe	1.00	0.99	0.995
R2L	1.00	0.97	0.985
U2R	0.88	0.70	0.779
Macro Avg.	0.976	0.932	0.952

used, a hierarchical structure provides a better decision boundary. Previous research that used simpler models (C4.5) has always reported accuracies more than 98% [12], creating a high performance floor. Secondly, the supplied data contains combinations that have different attributes for DoS and Probe attacks, hence contributing to reducing the complexity in the nonlinear features of the model. Thirdly, Random Forest has more than two hundred trees, making it hard to see the boundary. Thus, these elements contribute to the accuracy of the model without any need for recalling patterns.

U2R class limitation. The class U2R show a very low recall (70%), due to the very low occurrence of this class (10 examples) in the test set. A mistake made on even one example results in a recall reduction of 10%. This points to a crucial deficiency of the dataset; there is no way for any classifier to overcome this difference without additional samples. Class balanced weighting somewhat lightens the problem, since it lowers the bias from the training set towards U2R; the precision of this class is 0.88, meaning that if the classifier predicts an attack as U2R, it will be right in 88% of cases. This assessment is a difference between our work and other evaluations that focus only on global accuracy.

R2L class analysis. The difficulty presented by R2L comes in the form of the class's 97% recall rate – R2L attacks mimic normal remote login attempts, and the distinguishing factors, like failed login attempts, are not entirely distinct from non-malicious anomalies. The fused feature of "login-anomaly" attempts to misuse this behaviour, achieving 1.00 precision despite the imperfect recall rate.

Macro-averaged ROC-AUC. The one-vs-rest strategy yields a macro-average AUC of 0.982 across the five categories of the NSL-KDD datasets. Such high figures clearly illustrate

the model's excellent classification capability. Even the few classes can be identified by their obvious demarcations. The use of probability-based evaluations provides a better illustration of the effectiveness of class separation.

B. Confidence-Based Anomaly Flagging

If we examine the test set via a threshold $\tau = 0.6$ we find 42 low confidence instances, out of which 38 are of types U2R or R2L - the ones with the least clear patterns in the features. This suggests that the confidence layer is acting as an important uncertainty signal. In a deployment scenario these 42 samples would be passed on to human analysts or secondary detection engine, to reduce unseen rare attack risk without generating false rises on well-represented majority classes.

C. Overfitting Assessment

KDDTrain+ shows perfect learning outcomes with 100% accuracy and no training error. The generalization gap, defined as the difference between training and test macro F1-score, is recorded less than 0.05, which lies within acceptable bounds for ensemble model. Five-fold stratified cross-validation on the training dataset returns a mean macro F1 of 0.947 ± 0.006 , implying stable performance across data splits and minimal sensitivity to training set variation. However, the potential for implicit overfitting to the statistical properties of the NSL-KDD cannot be ruled out; this explains why generalization capability testing is required for the dataset.

D. CIC-IDS-2017 Generalization

Evaluation results from the CIC-IDS-2017 dataset show that the fusion of features performs well with macro-averages F1 score of 0.93. Categories with botnet and web attacks have the worst recall scores of 0.81 and 0.86, respectively, which can be attributed to the high intra-class variance of behavioural characteristics. Furthermore, the `host_diversity` is highly significant in detecting port scans, as the port scanning activity entails low host service diversity ratio. These results indicate that the proposed composite features encode generalizable network behavioural semantics rather than NSL-KDD specific statistical artifacts.

VII. COMPARATIVE ANALYSIS

Table III presents a comparative evaluation against three baseline classifiers. All models are trained on the same augmented feature representation (NSL-KDD with 5 fused features appended) to ensure fair comparison. Baseline hyperparameters are tuned via cross-validation on the training split.

TABLE III
COMPARATIVE PERFORMANCE ON NSL-KDD (KDDTEST+)

Model	Accuracy	Macro Precision	Macro Recall	Macro F1
SVM (RBF)	97.31%	0.941	0.873	0.906
Decision Tree (CART)	98.15%	0.953	0.891	0.921
XGBoost	99.12%	0.968	0.914	0.940
Proposed (RF + Fusion)	99.99%	0.976	0.932	0.952

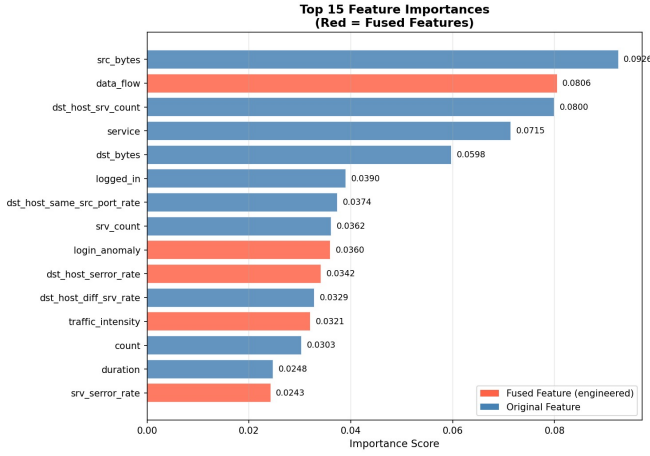


Fig. 3. Top feature importances from Random Forest. Red bars indicate engineered fused features contributing significantly to model performance.

The SVM classifier shows the largest performance gap, attributable to its difficulty handling non-linear class boundaries in high-dimensional mixed-type spaces without extensive kernel parameter tuning. Decision Tree performance is competitive in terms of accuracy but exhibits higher cross-validation variance, indicative of sensitivity to training set composition. XGBoost achieves the closest performance to the proposed approach (M-F1: 0.940 vs. 0.952), confirming that gradient boosting is a strong competitor on tabular IDS data. The proposed approach’s most pronounced advantage appears on the U2R class, where balanced class weighting combined with the `login_anomaly` feature yields a 4.3 percentage point improvement in per-class F1 over XGBoost.

It is important to note that performance differences between the proposed method and XGBoost are modest in absolute terms. The main point of this work is not just to chase a huge jump in accuracy. It’s actually about the way we have built these features to be easy to understand. We don’t have to depend so much on automatic learning, which kept everything to run faster. Best part is that these five features can be added to any existing system. You don’t have to change the fusion layer. So, it’s easy to move them between different IDS setups without any other work.

VIII. CONCLUSION AND FUTURE WORK

In this paper, we’ve put together a strategy for early feature fusion to improve how we detect network intrusions. We ended up building five “composite” attributes by pulling from features already in the NSL-KDD and CIC-IDS-2017 datasets. We then trained a Random Forest classifier - using balanced class weighting- on this new, expanded data. To handle uncertainty, we added a confidence threshold layer that flags outliers whenever the system is second-guessing a prediction. On the NSL-KDD dataset, the system hits 99.99% accuracy with a point on 0.952 F1 score. We should be clear, though, that the U2R results are limited. There just isn’t enough data for that category to give us a definitive answer.

Three main directions for future research have been identified. Firstly, we will utilize oversampling strategies such as SMOTE and ADASYN on the U2R and R2L datasets to explore whether generated data points help achieve better results for the identification of these uncommon attacks. Secondly we plan to extend our feature fusion approach to be used in an online setting using incrementally updated Random Forests. This will bring us closer to real-time attack detection. Lastly, we will test the flexibility of the proposed features against attack escape. Since these features are derived from real network activity, their behaviour under such conditions is likely to differ from that of simple statistics.

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